

Supplementary Information for:
ChEmbed: Enhancing Chemical Literature Search Through
Domain-Specific Text Embeddings

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Section 5.3 of the main text presents two related diagnostic analyses. The first asks whether chemistry literature and chemistry-related encyclopedic text are already distinguishable in the unmodified base-model space; the second examines how fine-tuning changes the representations of the same passages. This supplement describes the design and evidence supporting both parts of that argument.

S1 Sampling and encoding

We analyzed passages from the three retrieval benchmarks used in the main paper: ChemRxiv Retrieval, ChemNQ Retrieval and ChemHotpotQA Retrieval. To make the sampled passages independent at the source-document level, we grouped ChemRxiv passages by preprint and grouped ChemNQ and ChemHotpotQA passages by their source encyclopedia page, then retained one passage from each group. The primary comparison contained 800 ChemRxiv paragraphs and an equally sized encyclopedic sample composed of 400 ChemNQ and 400 ChemHotpotQA passages; sampling used seed 1337.

All three models received the same 1,600 passages after identical text preprocessing. Each input began with the model’s document-search prefix (`search_document:`). The encyclopedic benchmark records store the source-page title separately from the passage body, so this existing title field was placed before the passage text, as in the retrieval evaluation. ChemRxiv records have no separate title field and were therefore encoded using the paragraph text alone. The resulting 768-dimensional embeddings were L2-normalized, and inference was performed on CPU in single precision. We use ChEmbed_{vanilla} as the primary adapted model because it isolates the effect of fine-tuning from the tokenizer augmentation discussed in Section 5.2 of the main text, while ChEmbed_{progressive} provides a complementary sensitivity comparison.

S2 Separation before fine-tuning

The three corpora all concern chemistry, but they differ in genre, provenance, formatting and possibly in their topic distributions. We therefore asked a narrower question: before any ChEmbed fine-tuning, does the base model represent the literature and encyclopedic samples as measurably different populations?

To answer this question, we reduced the base-model embeddings to 64 principal components, standardized them and fitted logistic regression under stratified five-fold cross-validation. Dimensionality reduction and scaling were repeated inside each training fold, preventing information from the held-out passages from entering the classifier. The resulting out-of-fold accuracy was 95.6% (95% bootstrap interval 94.6–96.6%); because the two classes were balanced, uninformative predictions would be correct only about half of the time. To test whether the observed accuracy could arise by chance, we repeated the analysis 999 times after randomly shuffling the literature and encyclopedic labels. None of these shuffled analyses reached the observed accuracy, giving $p = 0.001$. Changing the projection size or cross-validation seed produced

accuracies between 94.9% and 96.1%. This use of classification as a two-sample test follows the general framework of Lopez-Paz and Oquab¹.

Passage length contributed to the distinction without accounting for most of it. A probe based only on word and character counts reached 76.1% accuracy, reflecting the greater average length of the preprint paragraphs. We therefore matched literature and encyclopedic passages within ten passage-length bands, leaving 766 documents with comparable length distributions. On this matched subset, a classifier using only word and character counts performed at chance (47.1%), while the embedding-based classifier remained accurate (93.6%). A local analysis led to the same conclusion: among the ten nearest neighbors of each passage, 86.4% came from the same corpus class, rather than the approximately even mixture expected if the classes were interchangeable².

Figure S1 shows this result directly. PCA selected the two directions that capture the most variation among the passage embeddings without reference to the corpus labels. Although these two directions contain only a small part of the total variance, they already reveal substantial separation.

Together, the classifier, length-matched control and neighborhood analysis show that the two corpora differ in their base-model representations. They do not identify the source of that difference, which may combine genre, topic, formatting and collection provenance.

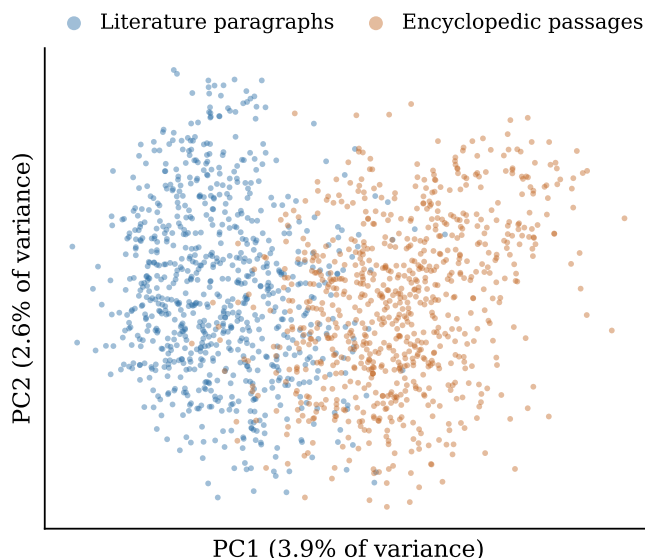


Figure S1. Literature and encyclopedic passages in the base model, projected onto their first two principal components. PCA calculated the projection from the embeddings alone, and the corpus colors were added afterward. Although the two components contain only a small share of the total variance, the classes already occupy substantially different regions.

S3 Changes from fine-tuning

We next encoded the same passages with the adapted models and compared the resulting spaces with the base representation. The clearest change concerned angular dispersion. For ChemRxiv, mean pairwise cosine similarity fell from 0.570 in the base model to 0.030 under ChEmbed_{vanilla}, whereas the encyclopedic pool changed from 0.517 to 0.163. Mean similarity therefore decreased by 0.540 for ChemRxiv and by 0.354 for the encyclopedic pool: the ChemRxiv decrease was 0.187 greater (95% bootstrap interval 0.180–0.195). ChEmbed_{progressive} produced the same qualitative pattern. Fine-tuning therefore dispersed both corpora, more strongly for the literature passages (Figure S2). This measure describes the concentration of the embedding space; on its own, it does not show that individual passages became easier to retrieve³.

The change in local structure was more modest and depended on the scale of the neighborhood. For each passage, we identified its ten closest passages before fine-tuning and then counted how many remained among its ten closest passages after fine-tuning. Under ChEmbed_{vanilla}, the average overlap was 35.4% for ChemRxiv and 42.7% for the encyclopedic passages, indicating somewhat greater local change for ChemRxiv. This difference disappeared when the neighborhood was expanded from ten to fifty passages (44.0% against 43.9%), while the corresponding ChEmbed_{progressive} comparison reversed direction at the wider scale (49.7% against 46.0%). Linear centered kernel alignment agreed with the ten-neighbor result⁴, but we did not use it for inference because ordinary bootstrap resampling was strongly biased at this sample size. These results support a graded, scale-dependent specialization rather than a wholesale reorganization confined to chemistry literature.

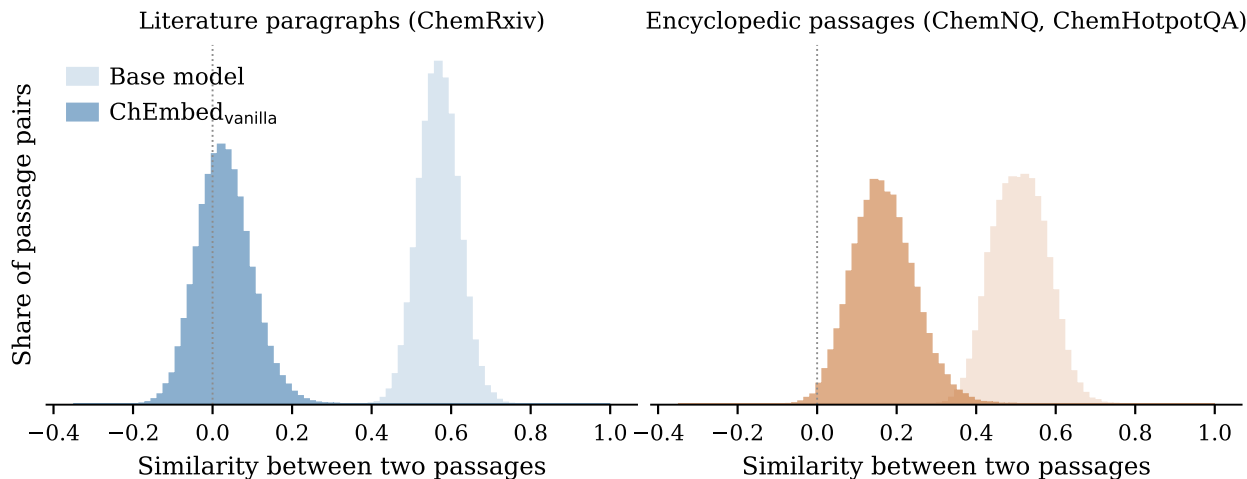


Figure S2. Distributions of pairwise cosine similarity before and after fine-tuning with ChEmbed_{vanilla}. Mean similarity falls from 0.570 to 0.030 for the ChemRxiv literature passages and from 0.517 to 0.163 for the balanced encyclopedic pool.

S4 Limitations

These analyses are observational: they establish that the sampled corpora are distinguishable in the base-model space and that fine-tuning changes their representations to different degrees, but they neither isolate the corpus property responsible for the separation nor demonstrate that the geometric changes caused the retrieval results. The geometry analysis also relies on one sampling seed, and the two encyclopedic retrieval tasks contain very few test queries. Their associated score differences should therefore be interpreted as indicative rather than conclusive.

References

- [1] D. Lopez-Paz and M. Oquab, International Conference on Learning Representations (ICLR), 2017.
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- [3] K. Ethayarajh, Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2019, pp. 55–65.
- [4] S. Kornblith, M. Norouzi, H. Lee and G. Hinton, Proceedings of the 36th International Conference on Machine Learning (ICML), 2019, pp. 3519–3529.